9 Equations and Formulas

9.1 Catalan Numbers

$C_n = \frac{1}{n+1}\binom{2n}{n}$

$C_0 = 1,\ C_1 = 1, \quad C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$

9.2 Other Combinatorial Identities

$\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$

$\sum_{i=0}^{k}\binom{n+i}{i} = \sum_{i=0}^{k}\binom{n+i}{n} = \binom{n+k+1}{k}$

$n, r \in \mathbb{N},\ n > r,\ \sum_{i=r}^{n}\binom{i}{r} = \binom{n+1}{r+1}$

If $P(n) = \sum_{k=0}^{n}\binom{n}{k}\cdot Q(k)$, then
$Q(n) = \sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}\cdot P(k)$

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9.3 Different Math Formulas

Pick's Theorem: $A = i + \frac{b}{3} - 1$

Derangements: $d(i) = (i-1)(d(i-1) + d(i0))$

$$\frac{n}{ab} - \left\{\frac{b'n}{a}\right\} - \left\{\frac{a'n}{b}\right\} + 1$$

9.4 GCD and LCM

If $m \in \mathbb{Z}$, then $\gcd(a+mb, b) = \gcd(a, b)$

If $\gcd(a_1, a_2) = 1$, then $\gcd(a_1 a_2, b) = \gcd(a_1, b)\cdot \gcd(a_2, b)$

$\gcd(a, \operatorname{lcm}(b, c)) = \operatorname{lcm}(\gcd(a, b), \gcd(a, c))$

$\operatorname{lcm}(a, \gcd(b, c)) = \gcd(\operatorname{lcm}(a, b), \operatorname{lcm}(a, c))$

$\gcd(n^a - 1, n^b - 1) = n^{\gcd(a,b)} - 1$

$\gcd(a, b) = \sum_{k|a,k|b}\phi(k)$

$\sum_{k=1}^{n}[\gcd(k, n) = 1] = \phi(n)$

$\sum_{k=1}^{n}\gcd(k, n) = \sum_{d|n} d\,\phi(n/d)$

$\sum_{k=1}^{n} x^{\gcd(k,n)} = \sum_{d|n} x^d\,\phi(n/d)$

$\sum_{k=1}^{n}\frac{1}{\gcd(k,n)} = \sum_{d|n}\frac{1}{d}\cdot\phi(n/d)$

$\sum_{k=1}^{n}\frac{k}{\gcd(k,n)} = \frac{n}{2}\sum_{d|n}\frac{1}{d}\cdot\phi(n/d)$

$\sum_{k=1}^{n}\frac{n}{\gcd(k,n)} = 2\sum_{k=1}^{n}\frac{k}{\gcd(k,n)} - 1$ (for $n > 1$)

$\sum_{i=1}^{n}\sum_{j=1}^{n}[\gcd(i, j) = 1] = \sum_{d=1}^{n}\mu(d)\left\lfloor\frac{n}{d}\right\rfloor^2$

$\sum_{i=1}^{n}\sum_{j=1}^{n}\gcd(i, j) = \sum_{d=1}^{n}\phi(d)\left\lfloor\frac{n}{d}\right\rfloor^2$

$\sum_{i=1}^{n}\sum_{j=1}^{n} ij\cdot[\gcd(i, j) = 1] = \sum_{i=1}^{n}\phi(i)\cdot i^2$

$F(n) = \sum_{i=1}^{n}\sum_{j=1}^{n}\operatorname{lcm}(i, j)$

$= \sum_{l=1}^{n}\left(\frac{(1+\lfloor n/l\rfloor)\lfloor n/l\rfloor}{2}\right)^2\sum_{d|l}\mu(d)ld$

9.5 Trigonometry

$\sin^2 x + \cos^2 x = 1$

$\tan x = \frac{\sin x}{\cos x}$

$\sin(2x) = 2\sin x\cos x,\ \cos(2x) = \cos^2 x - \sin^2 x$

$\sin(3x) = 3\sin x - 4\sin^3 x,\ \cos(3x) = 4\cos^3 x - 3\cos x$

$\sin(a \pm b) = \sin a\cos b \pm \cos a\sin b$

$\cos(a \pm b) = \cos a\cos b \mp \sin a\sin b$

Sum to product: $\sin a + \sin b = 2\sin\frac{a+b}{2}\cos\frac{a-b}{2}$

Product to sum: $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

Area of triangle: $A = \frac{1}{2}ab\sin C$

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$

Law of Tangents: $\frac{a-b}{a+b} = \frac{\tan\frac{A-B}{2}}{\tan\frac{A+B}{2}}$

9.6 Inverse Trigonometry

$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$

$\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$

$\tan^{-1} x = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

$\cot^{-1} x = \cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$

$\sin^{-1} x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$

$\cos^{-1} x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

$\sin^{-1} x = \cos^{-1}\sqrt{1-x^2}$

$\cos^{-1} x = \sin^{-1}\sqrt{1-x^2}$

9.7 Geometric Formulas

Triangle: $A = \frac{1}{2}bh$

Heron: $A = \sqrt{s(s-a)(s-b)(s-c)},\ s = \frac{a+b+c}{2}$

Circle: $A = \pi r^2,\ C = 2\pi r$

Rectangle: $A = lw$; Square: $A = a^2$

Trapezium: $A = \frac{1}{2}(a+b)h$

Regular n-gon: $A = \frac{1}{2}nR^2\sin\left(\frac{2\pi}{n}\right)$

Cube: $V = a^3,\ S = 6a^2$

Cuboid: $V = lwh,\ S = 2(lw + lh + wh)$

Cylinder: $V = \pi r^2 h,\ S = 2\pi r^2 + 2\pi rh$

Cone: $V = \frac{1}{3}\pi r^2 h,\ S = \pi r(r + l)$

Sphere: $V = \frac{4}{3}\pi r^3,\ S = 4\pi r^2$

Hemisphere: $V = \frac{2}{3}\pi r^3,\ S = 3\pi r^2$

Pyramid: $V = \frac{1}{3}(\text{Base area})\cdot h$

Frustum: $V = \frac{1}{3}\pi h(r_1^2 + r_1 r_2 + r_2^2),\ S_{\text{lateral}} = \pi(r_1 + r_2)l, l = \sqrt{(r_1 - r_2)^2 + h^2},\ S_{\text{total}} = S_{\text{lateral}} + \pi(r_1^2 + r_2^2)$

9.8 Number Theory

Divisor: $a \mid b \iff b = ak$

Prime: only $1, n$ divide $n > 1$

Unique factorization: $n = \prod_i p_i^{\alpha_i}$

Number of divisors: $\tau(n) = \prod_i(\alpha_i + 1)$

Sum of divisors: $\sigma(n) = \prod_i \frac{p_i^{\alpha_i+1}-1}{p_i-1}$

Product of divisors: $\mu(n) = n^{\tau(n)/2}$

n is perfect $\iff \sigma(n) = 2n$

$\pi(n) \approx \frac{n}{\ln n}$

Totient: $\phi(n) = n\prod_{p|n}\left(1 - \frac{1}{p}\right)$

$\sum_{d|n}\phi(d) = n$

ϕ is multiplicative: if $\gcd(m, n) = 1$, then $\phi(mn) = \phi(m)\phi(n)$

$\sum_{k=1}^{n}\phi(k) = \frac{n(n+1)}{2} - \sum_{i=2}^{n}\left\lfloor\frac{n}{i}\right\rfloor\phi(i)$

Fermat: $x^{p-1} \equiv 1 \pmod p$

Euler: $x^{\phi(m)} \equiv 1 \pmod m$

Wilson: $(n-1)! \equiv -1 \pmod n \iff n$ prime

Lagrange: $n = a^2 + b^2 + c^2 + d^2$

Zeckendorf: unique sum of non-consecutive Fibonacci numbers

Pythagorean triple: $(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2),\ \gcd(m, n) = 1$, one even

Goldbach, twin prime, Legendre (unproven conjectures)

Linear Diophantine: If $ax + by = c$ has solution (x_0, y_0), then
$(x, y) = \left(x_0 + \frac{b}{\gcd(a,b)}k,\ y_0 - \frac{a}{\gcd(a,b)}k\right),\ k \in \mathbb{Z}$

Example: $x_0 = 8,\ y_0 = -20$

9.9 Binomial Coefficient Identities

Definition: $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Symmetry: $\binom{n}{r} = \binom{n}{n-r}$

Pascal's Rule: $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$

Boundary: $\binom{n}{0} = \binom{n}{n} = 1$

Multiplicative: $\binom{n}{r} = \dfrac{n(n-1)\cdots(n-r+1)}{r!}$

Summation: $\displaystyle\sum_{r=0}^{n} \binom{n}{r} = 2^n$

Vandermonde's Identity: $\displaystyle\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$

Binomial Theorem: $(a+b)^n = \displaystyle\sum_{r=0}^{n} \binom{n}{r} a^{n-r}b^r$

9.10 Polygon Formulas

Sum of interior angles: $S = (n-2)\cdot 180^\circ$

Each interior angle (regular n-gon): $\theta = \dfrac{(n-2)\cdot 180^\circ}{n}$

Sum of exterior angles: 360°

Each exterior angle (regular n-gon): $\dfrac{360^\circ}{n}$

Number of diagonals: $D = \dfrac{n(n-3)}{2}$

Perimeter (regular n-gon, side length a): $P = n \cdot a$

Area (regular n-gon, circumradius R): $A = \dfrac{nR^2}{2}\sin\left(\dfrac{2\pi}{n}\right)$

Area (regular n-gon, side length a): $A = \dfrac{na^2}{4\tan(\pi/n)}$

Apothem (regular n-gon, side length a): apothem $= \dfrac{a}{2\tan(\pi/n)}$

Side length from circumradius: $a = 2R\sin\left(\dfrac{\pi}{n}\right)$

Circumradius from side length: $R = \dfrac{a}{2\sin(\pi/n)}$